COMPUTER FUNDAMENTALS

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MODULE I — FUNDAMENTALS OF COMPUTER SYSTEM

■ Meaning and Characteristics of Computer

A **computer** is an electronic device that accepts raw data from the user as **input**, processes it according to a set of instructions (the program) stored inside it and finally gives meaningful information as **output**. The word computer comes from the word “compute”, which means to calculate. A computer can store a huge amount of data and can also recall any part of it within a fraction of a second.

The important **characteristics and features** of a computer are:

1. **Speed** — A computer works with a speed of millions of operations per second (measured in MHz / GHz).
2. **Accuracy** — If correct input and correct program are given, the computer always produces a correct result; errors occur only because of wrong data (GIGO — Garbage In, Garbage Out).
3. **Diligence** — A computer never gets tired, bored or loses concentration like a human being, and can work continuously for hours together.
4. **Versatility** — The same machine is used for preparing documents, playing media, calculations, communication, banking, designing and entertainment.
5. **Powers of remembering (storage)** — It can store any amount of information for further use, and can recall it whenever required.
6. **Automation** — Once a program is given, the computer completes the whole task automatically without human interference.
7. **No I.Q.** — A computer cannot think and take its own decisions; it works only on the instructions given by the user.

> **Remember :** A computer does not have its own intelligence. It only follows the stored program; the quality of output depends completely on the quality of input and program given.

■ Components and Organization of Computer

A computer is organised in an arrangement of functional units. Every task passes through these basic units:

1. **Input Unit** — accepts data and instructions from the user through devices like keyboard, mouse and scanner, and converts them into binary form for the machine.
2. **Central Processing Unit (CPU)** — the brain of the computer; it is again divided into (a) **Arithmetic and Logic Unit (ALU)** which performs all calculations and logical comparisons, (b) **Control Unit (CU)** which

co-ordinates all the units and controls the flow of data, and (c) **Registers** which are very fast temporary storage inside the CPU.

3. **Memory Unit** — stores the program, data and intermediate results. It has **primary memory** (directly addressable by the CPU) and **secondary memory** (permanent storage).
4. **Output Unit** — converts machine results back into human-readable form through devices such as monitor and printer.

All the above units communicate with each other through three types of paths called buses: the **address bus** (carries the address of the location), the **data bus** (carries the actual data) and the **control bus** (carries control signals from the control unit). This whole design, where the program and the data are kept together in one memory and executed by one processor, is called the **stored program concept (von Neumann architecture)**.

■ Computer Generations and Classification

A generation is a step of technological improvement in computer manufacturing. First to fifth generations are:

Generation	Period	Technology used	Characteristics / Example
First	1940–56	Vacuum tubes	ENIAC, EDVAC; huge size, enormous heat, punched cards
Second	1956–63	Transistors	Smaller, faster, reliable; IBM 1401; assembly language
Third	1964–71	Integrated Circuits (IC)	IBM 360; keyboard and monitor, OS concept
Fourth	1971–present	Microprocessor (LSI, VLSI)	Personal computers; advanced GUI, networking, Internet
Fifth	Present & future	ULSI + Artificial Intelligence	Parallel processing, natural language, robotics

Classification of computer: On the basis of working principle they are **analog** (measures continuous quantities like temperature, speed), **digital** (works on 0 and 1) and **hybrid** (combination of both). On the basis of size the classes are **micro computer** (PC, laptop), **mini computer** (medium multi-user system), **mainframe** (large organisation records) and **super computer** (scientific research, weather forecasting). The syllabus specifically adds **distributed computers**, in which work of one program is spread and shared over several computers joined in a network, and **parallel computers**, in which many processors execute different parts of the same task simultaneously to achieve great speed.

■ Computer Memory — Memory Cell and Organization

Memory is that part of the computer where data, program and results are kept. It is made of a very large number of **memory cells**. Each cell can store one bit (0 or 1) and every cell has a unique **address**, by which the CPU locates it. Eight bits together form a byte, and bytes are grouped into kilobyte (1024 bytes), megabyte, gigabyte

and terabyte. Memory hierarchy starts from registers and cache (fastest and costly), then primary memory, then secondary storage. As we go down the hierarchy the capacity increases but the speed and cost per byte decrease.

■ Types of Memory — Primary and Secondary

Primary memory (main memory) is directly accessible to the CPU and is small and costly:

- **RAM (Random Access Memory)** — read and write memory; it is **volatile**, i.e. its contents are lost when power is switched off. Types: SRAM (fast, used for cache) and DRAM (dense, needs refreshing).
- **ROM (Read Only Memory)** — permanent memory written by manufacturer; stores boot programs. It is non-volatile.
- **PROM (Programmable ROM)** — once blank; the user can write it only once with a special device.
- **EPROM (Erasable PROM)** — its contents can be erased with ultra-violet light and then re-programmed with electrical signals.
- **EEPROM / Flash** — can be erased electrically, block by block; used in BIOS chips and pen drives.

Secondary storage is external, non-volatile, cheap and of very high capacity, but slower. Devices per the syllabus are **Floppy Disk (FD)** — old magnetic removable disk of 1.44 MB; **CD** — optical disk of about 700 MB written by laser; **HDD (Hard Disk)** — a stack of magnetic platters, the main storage of a computer; **Pen drive** — flash memory device connected to the USB port; **DVD** — digital versatile disk of 4.7 GB or more; and **Tape drive / DAT (Digital Audio Tape)** — sequential magnetic storage used mainly for taking backup of large data. For every primary memory type the student should remember the corresponding advantage and disadvantage, e.g. RAM is fast but volatile, ROM is reliable but read-only, EPROM is re-usable but costly.

■ Input Devices

Main general devices: keyboard (typewriter type keys), mouse (moving cursor on screen). The following special devices are listed in the syllabus:

- **Touch screen** — a display which is also an input device; the user directly presses on it (used in ATMs, mobiles and kiosks).
- **OMR (Optical Mark Reader)** — senses dark marks placed in pre-defined positions on a sheet; used for checking MCQ answer sheets and surveys.
- **OVR (Optical Voice Reader)** — reads voice input for recognition purposes.
- **OCR (Optical Character Reader)** — scans printed / typed characters of documents and converts them into editable computer text.
- **Light Pen** — a pointing device which is directly touched on the screen of a CRT monitor to draw or select.
- **Scanner** — optically scans a photograph, book page or document and digitizes it into the memory of computer.

■ Output Devices

Output devices present the processed information to the user, mostly on screen or on paper:

1. **Digitizers (graphics tablet)** — a flat pressure-sensitive pad which gives very high precision for design and CAD work.
2. **Plotters** — produce large hard-copy drawings of engineering graphs, maps and building plans on big sheets of paper using pens (drum plotter / flatbed plotter).
3. **LCD (Liquid Crystal Display)** — flat, light, low power liquid crystal screen used in modern monitors and laptops.
4. **Plasma Display** — large flat panel display using ionised gas cells; it gives bright picture over large area.
5. **Printers** — take output on paper. **Impact printers** strike a ribbon over paper (daisy wheel, drum, dot-matrix printer), while **non-impact printers** work without striking (ink-jet printer, laser printer, thermal printer).

MODULE II — PROCESSOR, INTERNET AND WORLD WIDE WEB

■ Processor — Structure of Instruction and Description

The whole operation of a computer is decided and performed by the processor. An **instruction** has two parts: **Opcode (operation code)**, which tells what operation is to be performed, and **operand**, which tells on what data or where the data is stored. For example, in the instruction ADD A, B — ADD is the opcode and A, B are the operands.

The working of the processor is a repeated cycle called the **instruction cycle**: Fetch (bring the instruction from memory) → Decode (interpret the operation) → Execute (perform the operation) → Store the result back. **Processor features** on which its power is measured are: **word size** (number of bits handled at a time, e.g. 32-bit / 64-bit), **clock speed** (frequency expressed in GHz), **size of cache and registers**, **addressing capability** and the length and richness of the instruction set.

■ RISC and CISC

Processors are designed on two philosophies — **CISC (Complex Instruction Set Computer)** and **RISC (Reduced Instruction Set Computer)**. **CISC** keeps a large set of complex, variable-length instructions; one instruction may perform many micro-operations; it needs fewer registers and the hardware decoding logic is complex. **RISC** keeps only a small set of simple, fixed-length instructions which mostly execute in one machine cycle; it uses a large number of registers and compiler-dependent pipelining, which makes the hardware simple and very fast. ARM and modern mobile processors are RISC based, while Intel x86 is CISC based.

■ Introduction to Internet and Internet Access

Internet is a world-wide network of millions of smaller networks connected with the help of telephone lines, optical fiber and satellites, which works on the TCP/IP set of protocols and provides services like e-mail, file transfer and the World Wide Web. **Internet access** means the way by which a user is connected to the Internet: available options are **dial-up telephone line** (slow, blocks the phone), **DSL / broadband cable** (always-on fast connection through modem), **leased line**, **wireless and mobile access** (3G / 4G / 5G, Wi-Fi, WiMAX) and **satellite access** for remote areas.

■ Protocols, Addressing, Web Pages and HTML

A **protocol** is the set of rules which both sides of a data communication follow. The most important protocols for the Internet are **TCP/IP** — the transmission rules of the whole network (TCP takes care of breaking data into packets and joining them reliably, IP takes care of the addressing and routing of every packet), **HTTP** — Hyper Text Transfer Protocol, that is the rules for transferring web pages, and **FTP** — File Transfer Protocol, which uploads and downloads files between machines.

Addressing: every machine on the Internet has a 32-bit / 64-bit numeric **IP address** (for example 172.16.254.1), and friendly names called **domain names** (like www.google.com) are used in the place of addresses which the **DNS server** resolves into IP numbers. **World Wide Web** is the collection of interlinked documents stored on web servers. **Web pages** are written in **HTML** (Hyper Text Markup Language) with tags and hyperlinks, reach the user from a web server and are displayed by a **web browser** such as Chrome, Firefox or Edge. Searching for information is done through **search engines** which accept keywords and give a list of matching sites.

■ Applications, Advantages and Disadvantages of Internet

Applications of Internet: e-mail and messaging, information and knowledge search, education (e-learning), online banking, shopping (e-commerce), entertainment, social media, chats and online meetings.

Advantages: world-wide and very fast communication, huge central information source available 24 hours, easy sharing of data and resources, cheap and paperless services. **Disadvantages:** theft of personal information, viruses and hackers, spreading of false information, wastage of time and harmful content for children; in addition, security of data and privacy of users becomes a constant concern.

MODULE III — NUMBER SYSTEM AND ARITHMETIC

■ Decimal, Octal, Hexadecimal and Binary Number Systems

A number system is a way of representing numbers with a fixed base. The **base** is the number of different symbols (digits) used by that system.

System	Base	Digits used	Example
Binary	2	0, 1	1011
Octal	8	0 – 7	745
Decimal	10	0 – 9	459
Hexadecimal	16	0 – 9 and A – F	2D8A

In hexadecimal, the values of letters are A = 10, B = 11, C = 12, D = 13, E = 14 and F = 15.

■ Conversions within and between the Number Systems

Decimal to Binary, Octal or Hexadecimal

Divide the decimal number repeatedly by the target base and write the remainders from bottom to top. For the fractional part, multiply the fraction by the base repeatedly and read the integer parts from top to bottom.

Binary, Octal or Hexadecimal to Decimal

Multiply every digit by the value of its position (copy base to the power of position, counted from the right starting with 0) and add all the products.

> $$For\ binary\ 1011 : 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 = 8 + 0 + 2 + 1 = 11\ in\ decimal$$

■ Binary Arithmetic and Complement Method

Binary addition rules are the same as the rules of decimal addition except that the carry is 1 as soon as the sum becomes two. 1 + 0 = 1, 1 + 1 = 10 (0 with carry 1), 1 + 1 + 1 = 11 (1 with carry 1).

Binary subtraction is done with the help of the **complement method** per the syllabus. The **1's complement** of a binary number is found by writing 0 in place of every 1 and 1 in place of every 0. The **2's complement** is obtained by adding 1 to the rightmost digit of the 1's complement.

Subtraction using 2's complement: to find A − B, take the 2's complement of B and add it to A. If a final carry is produced, discard that carry and the answer is positive. If no carry is produced, then take the 2's complement of the answer and put a minus sign before it — that is the required result. The same rule works for octal, hexadecimal and decimal with their own base complements.
